

Debugging Log

Entry 1

13 minutes spent debugging

Failure

Not Compile: type checker complains: Object is possibly undefined

Experiments

Experiment 1.1

Question: How to suppress the warning: adjacent.get(startStr) is possibly undefined (adjacent is a Map)

Steps Taken: throw error when startStr is false, and set(startStr, []) when adjacent.has(startStr) is false.

Result: the type checker still complains

Lesson: I need to directly claim it's impossible to be undefined, or directly throw error if it is undefined to make the type checker believe

Error

```
31: if (!adjacent.has(startStr)) { adjacent.set(startStr, []); }
```

I do not have an if statement to throw error when adjacent.get(startStr) is undefined.

Defect

```
34: adjacent.get(startStr).push(edge);
```

The type checker believes the get function might return undefined, and complains.

Debugging Scope

This failure **was** in the same function as the bug. To find the bug, I debugged through the following functions:

```
server/src/dijkstra shortestPath
```

Entry 2

1.2 hours spent debugging

Failure

Error fetching /api/shortestPath: 200 response is not valid JSON

Experiments

Experiment 2.1

Question: where does the failure occur?

Steps Taken: Trace the error log and locate the problem

Result: error log: "200 response is not valid JSON", means the response

Lesson: The defect comes from parsing the json

Experiment 2.2

Question: What is the bug in doShortestPathJson

Steps Taken: check what record does the server sends back

Result: The servers sends back a record including a single Path

Lesson: The servers sends a Path, but the "path" client want is Edge[], which is path.steps

Experiment 2.3

Question: The type checker complains data might be undefined in data.path.steps

Steps Taken: go through step by step to eliminate the possible type of data.path.steps

Result: finally, the type checker stops complaining

Lesson: I need to eliminate all possible undefined.

Error

```
169: this.setState({ path: data.path });
```

data.path is Path, but the "path" key is an Edge[], which is data.path.steps as specified in dijkstra.ts

Defect

```
148: .catch(() => this.doShortestPathError("200 response is not valid JSON"));
```

The mismatch of type of the record's key and value leads to the failure of json parsing

Debugging Scope

This failure **was not** in the same function as the bug. To find the bug, I debugged through the following functions:

```
client/src/App.tsx doShortestPathResp
server/src/routes.ts getShortestPath
server/src/dijkstra.ts Path
client/src/App.tsx doShortestPathJson
```

Logged a total of **1.4 hours** of debugging.

Edit Log